



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
THIRD SESSIONAL
SUBJECT: (CE 701) ARTIFICIAL INTELLIGENCE

Examination	: B.Tech Semester VII	Seat No.	:
Date	: 09-10-2023	Day	: Monday
Time	: 2.00 to 3.15 PM	Max. Marks	: 36

INSTRUCTIONS:

- Figures to the right indicate maximum marks for that question.
- The symbols used carry their usual meanings.
- Assume suitable data, if required & mention them clearly.
- Draw neat sketches wherever necessary.

Q.1 Do as directed.

[12]

CO3 R (a) What are advantages and disadvantages of semantic Nets?

[2]

CO3 E (b) Unify: $f(x, x)$ and $f(g(y), g(y))$

[2]

Write the output of unification if they unify, else mention reason if they don't unify.

CO3 U (c) How does Resolution work?

[2]

CO6 R (d) Explain Uniform Crossover with crossover mask with an example.

[2]

CO6 A (e) Generate soundex codes for following words. Substitutions for generating the soundex codes are listed in following table.

[2]

Words: network, Henary, Ashcraft, Water

Letter(s)	Substitute with Integer
B, F, P, V	1
C, G, J, K, S, X, Z	2
D, T	3
L	4
M, N	5
R	6

CO6 A (f) Consider following corpus:

[2]

<s> I am Trisha </s>

<s> I like school</s>

<s> Do Trisha like school </s>

<s> Trisha I am</s>

<s> Do I like school</s>

<s> Do I like Trisha</s>

<s> I do like Trisha</s>

What is the most probable next word predicted by the model for the following statement? Use Tri-gram.

<s> Do I like ?

[Note: ? means what will be the most probable word after like]

Q.2 Attempt Any TWO from the following questions.

CO3 N (a) Discuss Inheritance algorithm based on path length for tangled hierarchies.

[12]

CO3 N (b) Write Convert to Conjunctive Normal Form (Clause Form) algorithm, used by Resolution.

[6]

CO3 N (c) Define various Semantic Properties of WFF. Explain the term Model and Counter-Model with a suitable example.

[6]

Q.3 Attempt the following questions.

CO6 A (a) Consider following corpus:

[12]

<s> heavy rain in costal area</s>

[6]

<s> long rain today</s>

<s> having fun in heavy rain</s>

<s> rain seasons starts in costal area</s>

Which of the following sentence is more probable?

- $\langle s \rangle$ rain in costal area $\langle /s \rangle$
- $\langle s \rangle$ having fun in costal area $\langle /s \rangle$

Determine using **unigram and bigram** model. Show all necessary calculations.

CO6 N (b) Explain Rank-based Selection with an example. Compare Rank based selection with Roulette-Wheel Selection. [6]

OR

CO6 A (a) Which types of ambiguity is observed for each of the following sentence? Justify your answer. [6]

- I visited a bank.
- I saw the star with a telescope.
- Watch what you want, when you want.
- Parva and Nirva both are married.
- I went to hospital, and they told me to go home and rest
- Aajaayenge

CO6 N (b) Using genetic algorithm maximize $f(x) = (-x^2 / 10) + 3x$ over $\{0, 1, 2, \dots, 31\}$ initial x value of $\{11, 12, 14, 22\}$. Show at least three iteration of Genetic algorithm. [6]



DHARMSINH DESAI UNIVERSITY, NADIAD
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SUBJECT: (CE - 714) Image Processing

: B.Tech Semester VII

Seat No. :

: 10-10-2023

Day :

: Tuesday

: 02:00 PM To 03:15 PM

Max. Marks :

: 36

INSTRUCTIONS:

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Do as directed.

- Q-1 A (a) What structuring element is used for boundary extraction using morphological operation? [12]
CO2 A (b) What should be the structuring element to extract 3-pixel wide boundary. [2]
CO3 N (c) What is the entropy of the information source? How to measure it? How is it useful? [2]
CO2 R (c) For The following 3 bit image [2]

0	1	1	2
2	5	5	4
6	7	6	2
4	6	5	0

Between class variance for all possible thresholds is found as follow :

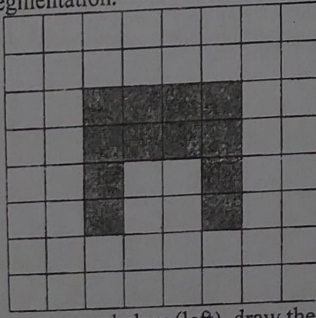
t	1	2	3	4	5	6
σ_b^2	1.5	2.5	7	3.5	2	1

Segment the given matrix using the above mentioned table.

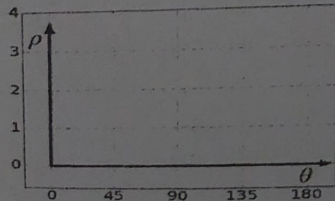
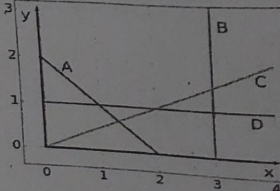
- CO4 U (d) Formulate the non maxima suppression scheme in canny detection for 3 X 3 region. [2]
CO8 E (e) Describe the working of block transform coding with an example. [2]
CO8 A (f) What are the various redundancies present in an image? Give examples of each. [2]

Attempt Any TWO from the following questions.

- Q2 CO4 C (a) Factors such as noise and nonuniform illumination play a major role in the performance of a thresholding algorithm. Discuss the techniques for choosing the thresholds in such situation. [12]
CO4 E (b) Discuss Region splitting and merging approach of region based segmentation. [6]
Consider a 8 x 8 image shown in figure below. Segment the image using the split and merge procedure. Let $Q(R_i) = \text{TRUE}$ if all pixels in R_i have the same intensity. Show quad tree corresponding to your segmentation.



- CO4 E (c) For each of the lines in the image space below (left), draw the corresponding point in the hough space (right). Use letters A, B, C and D to mark the points. You need to also write the point coordinates ($\arctan(2) \approx 64$ degrees). Assume $\rho \geq 0$ and $0 \leq \theta < 180$. [6]



Q.3

Attempt the following.

CO4

E

- (a) i. If you implement region filling using a structuring element containing all the eight neighbors of the origin what problem will you face? Illustrate with an example.
ii. Explain with the extraction of connected components using the morphological operations.

CO3

E

- (b) Show and explain the basic components of a lossless predictive coding system. Encode following data using the lossless predictive coding method using second order linear predictor (Average of previous 2 pixels). Also show the decoding of the encoded message.

21 21 21 21 95 169 243 243 243 243

OR

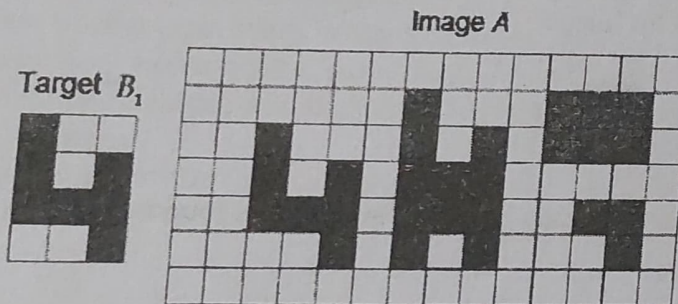
Q.3

Attempt the following.

CO4

U

- (a) Apply HIT or MISS transformation for the shape of B1 in image A.



CO3

E

- (b) A sequence is encoded using LZW algorithm and the initial dictionary shown in the table.

[06]

Index	Entry
1	A
2	B
3	R
4	T

The output of LZW encoder is the following sequence :

3 1 4 6 8 4 2 1 2 5 10 6 11 13 6 Decode this sequence.



DHARMSINH DESAI UNIVERSITY, NADIAD
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THIRD SESSIONAL
SUBJECT: (CE725) BIG DATA ANALYTICS

Examination
Date
Time

: B.Tech Semester VII
: 11-10-2023
: 2:00 PM to 3:15 PM

Seat No.
Day
Max. Marks

:
: Wednesday
: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
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4. Draw neat sketches wherever necessary.

Do as directed.

- Q1 R (a) Let suppose we have built the system using apache spark to perform real time analysis of tweets generated by twitter. The user will simply enter the list of twitter keywords to analyze (e.g. bitcoin). Tweets will be classified as positive, negative, or neutral based on analysis of the text. The tweet and sentiment results will be written to Cassandra or Hive. [12]

Enlist the apache spark components that can be used in order to implement the given application. Give a brief idea about the components. [2]

- Q2 U (b) Categorize the following transformations into Narrow transformation or Wide Transformation. [2]

- Q3 C (c) Let us consider that Flipkart is using apache spark accumulators in order to keep track of the delivered orders. On each worker node we have a variable called 'status' which gives you the current status of order. If the order is delivered then we have status as 'delivered'. [2]

- I. Create an accumulator on the driver node as 'delieverdaccum'.
- II. For each delivered order the accumulator should be updated such that the flipcart manager can observe the delivered orders.

- Q4 E (d) Explain algorithm followed by tag/word cloud generator. Convince the purpose of the Visualization phase of Big Data Analytics life cycle with it. [2]

- Q5 U (e) Discuss below: [2]
- (I) Data silos Vs Data Lakes
 - (II) Prescriptive Analytics Vs Cognitive Analytics.

- Q6 N (f) How is data stream analytics different from others? Illustrate different window types in windowing methods. [2]

Attempt Any TWO from the following:

- (a) Consider the following sequence database used in Generalized Sequential Pattern Mining (GSP): [12]

Sequence ID	Timestamp	Events
A	10	2, 3, 5
A	20	6, 1
A	23	1
B	11	4, 5, 6
B	17	2
B	21	7, 8, 1, 2
C	28	1, 6
	14	1, 8, 7

Formulate subsequences possible with minimum support = 50%. Prune subsequences upto length k ; where $k = 2$. Show only one entry of Join operation which results in the subsequence of length three (if any).

- CO3 (b) Deploy the solution for the given case study: NASA develops a climate model based on various factors affecting climate. NASA's Jet Propulsion Laboratory has developed a System which first loads the data, analyzes and transforms the climate data which is of the form (DATA includes latitude, longitude, time, value, height) and stores it into My SQL DB having 6 billion tuples resulting in overhead. Create a solution using APACHE HIVE architecture with graphical representation of steps involved in the solution.

- CO3 (c) Create the Pig Latin script for Word Count considering the following sentence: "This is a Hadoop Class, Hadoop is a Bigdata Technology." Show steps how to generate the same using Pig Latin.

Q.3 Attempt the following:

- CO5 N (a) The apache spark is configured with 12 nodes in a cluster, each node consists of 1 cores, each node will assign the memory of 128GB. if 3 cores can be used per ex out of all cores in worker nodes one core will be assigned to YARN or cluster manager. Find out the value for following properties for a spark cluster configuration.

- I. -- executor-cores
- II. -- num-executors
- III. -- executor-memory

- CO6 A (b) Do as Directed.

- I. Write a scala REPL to generate the RDD named as 'numberRDD' which contains odd numbers from a given list.
(34, 1, 25, 96, 1, 23, 56, 47, 49, 47, 65, 65, 8, 8, 11, 23, 101)
- II. Write a scala REPL to Remove duplicates from 'numberRDD'.
- III. Display total odd numbers in 'numberRDD'.
- IV. Write a scala REPL that finds the sum of all odd numbers generated by 'numberRDD'.
- V. Write the output of the sum of all odd numbers to the 'output' directory located in HDFS at path "/demo/number/location".
- VI. Show the DAG visualization for above operations with stages.

OR

Attempt the following:

- Q.3 CO5 N (a) What is the need of partitioning in apache spark? Enlist different types of partitioning techniques. Explain any one technique with suitable examples.
- CO6 A (b) Write a map reduce program using apache spark(Scala), which should count how many words belong to each of the following four length categories:
- tiny: 1 letter
 - small: 2-4 letters
 - medium: 5-9 letters
 - big: more than 10 letters

Bloom's Taxonomy levels : R-Remembering, U- Understanding, A-Applying, N-Analyzing, E- Evaluating, C-Creating

: Third Sessional
: 12/10/2023
: 02:00 PM to 03:15 PM

Seat No : _____
Day : Thursday
Max. Marks : 36

INSTRUCTIONS:

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2. Draw neat sketches wherever necessary.

Q.1 Do as directed.

- | | | |
|----|--|------|
| a) | Explain p-machine for PASCAL | [12] |
| b) | Explain Reduce-Reduce Conflict with suitable example | 3 |
| c) | What is the cost of using Basic Mark-and-Sweep collector vs. Baker's Mark-and-Sweep collector? | 3 |
| d) | Compare a synthesized attribute and an inherited attribute with an example of each. | 2 |
| e) | List the operations performed on a symbol table in increasing order of how frequently they are used. | 2 |

Q.2 Attempt Any Two

- | | | |
|----|---|------|
| a) | Consider the network of objects in Fig. C and write the reference counts of all nodes and state which nodes are garbage collected for the following situations:
[Note: all the 3 situations are <i>independent</i> of each other] | [12] |
| | 1) Node E is deleted | 6 |
| | 2) Link $G \rightarrow H$ changed to $G \rightarrow E$ | |
| | 3) Node C is deleted. | |

b) Show the status of

- 1) stack implemented a tree structured (use binary tree) symbol table
- 2) stack implemented a hash structured symbol table

Prior to the completion of compilation of block B2 for the segment of program given in **Fig. A**. Assume the size of the hash table is 4 and the addresses to be hashed are given in **Fig. B**.

- | | | |
|----|---|---|
| c) | Write a context free grammar to represent arithmetic expression involving +, * and = operators. Also, write a syntax directed definition to evaluate the given arithmetic expression. Draw annotated parse tree for sample input $a = (3 + 2) * 2 + 3$ and show evaluation of attributes. Write the evaluation order. Is this SDD S-Attributed? | 6 |
|----|---|---|

B1 { int a, x;

B2{ int a, b;

} int c, y;

Fig. A

Name	Address to be hashed
------	----------------------

a, b	1
c, x	2
y	3

Fig. B

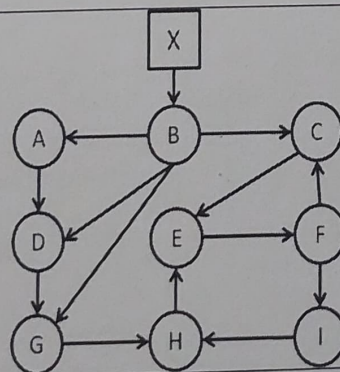


Fig. C

CO2-C Q.3 Answer the following [12]
Let $L = \{ a^i b^j c^m d^n \mid i=j \text{ and } m=n, j,n>0 \}$.
(a) Construct Canonical LR(0) itemset and also generate SLR(1) parsing table
(b) Perform Bottom-Up Parsing for the string : abccdd

OR

CO2-C Q.3 Answer the following [12]
(a) Construct Canonical LR(1) itemset for the grammar given below
(b) Generate LALR Parsing Table.

$S' \rightarrow S$
 $S \rightarrow A B$
 $A \rightarrow p A q$
 $A \rightarrow p q$
 $B \rightarrow q B r$
 $B \rightarrow q r$